

Housing Frictions, Spatial Misallocation, and Long-Run Growth

A Research Brief

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Motivation

Large productivity gains in modern economies are concentrated in a small number of high-density urban regions. At the same time, housing supply in these regions is often highly inelastic. This project studies how housing constraints restrict access to high-productivity cities, distort spatial allocation, and slow long-run economic growth.

The central question is:

How do housing frictions interact with agglomeration and innovation to shape aggregate growth dynamics?

Core Mechanism

The model formalises a simple mechanism linking housing markets to growth:

Housing frictions $\rightarrow$ Urban occupancy $\rightarrow$ Agglomeration and innovation $\rightarrow$ Long-run growth

Housing constraints limit effective access to high-productivity urban labour markets, reducing both static efficiency and the scale of innovation activities that drive frontier growth.

Model Overview

I develop a discrete-time growth framework with two accumulable inputs: physical capital and housing capital. Output is produced according to

$$Y_t = K_t^\alpha H_t^\beta (A_{\text{eff},t} L_t)^{1-\alpha-\beta}, \quad (1)$$

where housing capital affects production through its role in facilitating access to productive locations.

Total factor productivity evolves endogenously as a function of urban concentration, capturing the idea that dense economic environments foster knowledge spillovers and innovation. Housing frictions enter as constraints on urban occupancy rather than direct distortions to production technology.

Solution and Computation

The equilibrium is characterised implicitly and solved numerically. The solution approach combines:

- Steady-state calibration
- Fixed-point iteration for equilibrium conditions
- Forward simulation of transition dynamics

All numerical work is implemented in **MATLAB** and **Dynare**. The framework is designed to be modular, allowing counterfactual experiments on housing supply elasticity and urban access costs.

Preliminary Insights

Numerical simulations suggest that tighter housing constraints:

- Increase spatial misallocation by limiting access to high-productivity regions
- Reduce aggregate innovation intensity
- Slow convergence and lower long-run growth rates

These effects are quantitatively meaningful even when housing enters production only indirectly, highlighting the importance of spatial frictions for macroeconomic outcomes.

Current Status and Next Steps

Current progress

- Complete theoretical framework
- Working numerical solution pipeline
- Draft model note and literature review

Next steps

- Calibration using external housing and urban data
- Policy counterfactuals (housing supply reforms)
- Extension to heterogeneous regions

Selected References

- Acemoglu, D. (2009). *Introduction to Modern Economic Growth*.
- Glaeser, E. L., Gyourko, J., & Saks, R. (2005). Why Have Housing Prices Gone Up?
- Hsieh, C.-T., & Moretti, E. (2019). Housing Constraints and Spatial Misallocation.
- Lucas, R. E. (1988). On the Mechanics of Economic Development.

Full model details and replication materials available upon request.